

ASHTON DY

Knoxville, TN | (615) 524-0093 | ady@vols.utk.edu | [LinkedIn](#) | [Personal Website](#)
Honors Computer Science Student Exploring Computational Chemistry Research

EDUCATION

The University of Tennessee - Knoxville | *Tickle College of Engineering* Knoxville, TN
Bachelor of Science in Computer Science | **GPA: 3.98/4.0** Graduation: May 2027
Major: Honors Computer Science
Relevant Coursework: Chemoinformatics, Deep Learning, Reinforcement Learning, LLMs & Agents, Data Structures & Algorithms, Physical Chemistry, Probability and Random Variables

RESEARCH EXPERIENCE

Oak Ridge National Laboratory | *Research Student Intern* May 2026 – Present

- Developed and optimized visual molecular dynamics-based tools for membrane trajectory analysis.
- Cut down the membrane trajectory process time from 3 minutes to 5 seconds.
- Collaborated with peer interns to execute a GROMACS molecular dynamics workflow, analyzing membrane-protein systems and structural stability metrics.

Dr. Vogiatzis' Lab | *Undergraduate Research Assistant* May 2025 – Present

- Developed and fine-tuned a transformer-based molecule generation pipeline with multi-objective reinforcement learning.
- Secured collaborator to synthesize NLRP-3 inhibitor candidates.

Dr. Coble's Lab | *Undergraduate Research Assistant* September 2025 – Present

- Developed a multi-agent system to design sheet metal process sequences to optimize workflows.
- Programmed a comprehensive knowledge graph to provide structured insights into sheet metal processes.
- Designed and integrated knowledge graphs and vector databases into an LLM-driven RAG pipeline, leveraging Gemini and LangChain for enhanced retrieval and generation performance.

PROFESSIONAL EXPERIENCE

Clayton | *Software Engineer Intern* May 2025 – August 2025

- Spearheaded AI development by creating the first AI-powered internal tool.
- Researched and developed a robust Model Context Protocol based RAG pipeline for +10,000 employees.
- Utilized Amazon Web Services such as EC2, Lambda, Amazon Bedrock, and Amazon Kendra.

The University of Tennessee Housing | *Resident Assistant* August 2024 – August 2025

- Fostered a vibrant community of 58 upperclassmen through 13 engaging programs.
- Led individual meetings with 50+ residents in two weeks, strengthening their relationships and support.

INVOLVEMENT / LEADERSHIP

Society of Asian Scientists and Engineers | *Events Director & Outreach Director* January 2024 – Present

- Designed and organized 15 events for the full academic year, collaborating with other organizations.

Lambda Phi Epsilon | *Academic Chair & Web Chair* November 2026 – Present

- Developed and maintained website and organized academic success events

Tau Beta Pi | *Initiation Chair* March 2024 – Present

- Invitation-only engineering honors society that recognizes the top 12.5% of juniors and 20% of seniors.

TECHNICAL SKILLS

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- Programming Languages:** Python, C++, Javascript, Java, HTML, CSS, RISC-V Assembly
 - Technologies & Frameworks:** Pytorch, rdkit, pandas, MongoDB, Express.js, React.js, node.js, Tensorflow, SASS, Framer motion
 - Software & Tools:** Windows, Linux, GitHub, Microsoft Office, Google Suite, Docker, GROMACS

AWARDS / HONORS

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- University Honors, The University of Tennessee
 - Engineering Honors, Cook Grand Challenge
 - Summa Cum Laude Dean's List (Fall '23 – Spring '26)
 - Phi Kappa Phi – Top 7.5% of Juniors